CASE STUDY 7: EXAMINATION ANSWER EVALUATION

1. Problem Statement

Examination Answer Evaluation is used to calculate a student's marks by comparing the answers given by the student with the correct answer key. The program accepts the student's answers and the correct answers, compares them, calculates the total score, and displays the result.

2. Objectives

- To develop a Python program for evaluating examination answers.
- To compare student answers with the correct answer key.
- To calculate the total marks obtained.
- To display the student's score.
- To understand lists, loops, conditional statements, and counting in Python.

3. Assumptions

1. The examination contains five multiple-choice questions.
2. Each question carries one mark.
3. There is no negative marking.
4. Answers are entered using uppercase letters A, B, C, or D.
5. A correct answer earns one mark.
6. An incorrect answer earns zero marks.

4. Algorithm

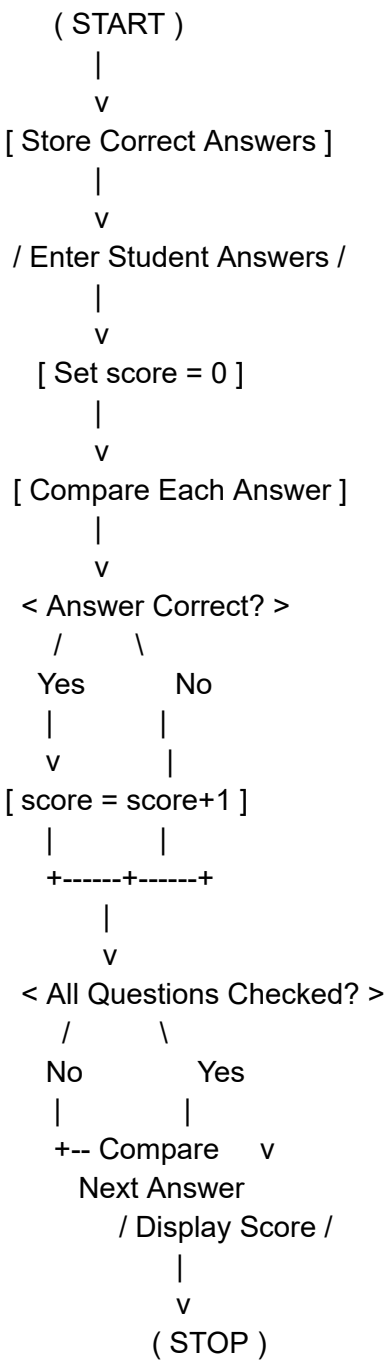
1. Start the program.
2. Store the correct answers in a list.
3. Accept the student's answers for five questions.
4. Initialize the score to zero.
5. Compare each student's answer with the corresponding correct answer.
6. If the answer is correct, increase the score by one.
7. Repeat the comparison for all five questions.
8. Display the total marks obtained.
9. Display the maximum marks.
10. Stop the program.

5. Flowchart

Flowchart symbols:

- Oval – Start/Stop
- Parallelogram – Input/Output
- Rectangle – Process
- Diamond – Decision
- Arrow – Direction of flow

Simple flowchart:



Draw Start and Stop using oval shapes, input/output using parallelograms, processing using rectangles, decisions using diamonds, and connect them with arrows.

6. Pseudocode

```
START

correct_answers = [A, B, C, D, A]
score = 0

INPUT student answers

FOR each question
   IF student answer = correct answer
      score = score + 1
   END IF
END FOR

DISPLAY score
DISPLAY maximum marks = 5

STOP
```

7. Python Source Code

```
correct_answers = ["A", "B", "C", "D", "A"]
score = 0

for i in range(5):
   answer = input("Enter answer for question " + str(i + 1) + ": ")
   answer = answer.upper()

   if answer == correct_answers[i]:
      score = score + 1

print("Marks Obtained:", score)
print("Maximum Marks: 5")
```

8. Test Cases and Expected Outputs

Test Case 1: All Answers Correct

Input:

```
Enter answer for question 1: A
Enter answer for question 2: B
Enter answer for question 3: C
Enter answer for question 4: D
Enter answer for question 5: A
```

Expected Output:

Marks Obtained: 5
Maximum Marks: 5

Test Case 2: Some Answers Correct

Input:

Enter answer for question 1: A
Enter answer for question 2: C
Enter answer for question 3: C
Enter answer for question 4: B
Enter answer for question 5: A

Expected Output:

Marks Obtained: 3
Maximum Marks: 5

Test Case 3: All Answers Incorrect

Input:

Enter answer for question 1: B
Enter answer for question 2: A
Enter answer for question 3: D
Enter answer for question 4: C
Enter answer for question 5: B

Expected Output:

Marks Obtained: 0
Maximum Marks: 5

Note: Run each test case separately. The correct answers are A, B, C, D, A.

9. Output Screenshots

```python
correct_answers = ["A", "B", "C", "D", "A"]
score = 0

for i in range(5):
    answer = input("Enter answer for question " + str(i + 1) + ": ")
    answer = answer.upper()

    if answer == correct_answers[i]:
        score = score + 1

print("Marks Obtained:", score)
print("Maximum Marks: 5")
```

```
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case7.
py
Enter answer for question 1: A
Enter answer for question 2: B
Enter answer for question 3: C
Enter answer for question 4: D
Enter answer for question 5: A
Marks Obtained: 5
Maximum Marks: 5
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case7.
py
Enter answer for question 1: A
Enter answer for question 2: C
Enter answer for question 3: C
Enter answer for question 4: B
Enter answer for question 5: A
Marks Obtained: 3
Maximum Marks: 5
>>>
= RESTART: C:/Users/sirit/OneDrive/Desktop/Python-Activity-Based-Learning/case7.
py
Enter answer for question 1: B
Enter answer for question 2: A
Enter answer for question 3: D
Enter answer for question 4: C
Enter answer for question 5: B
Marks Obtained: 0
Maximum Marks: 5
```

10. Conclusion

Thus, the Examination Answer Evaluation program was successfully implemented using Python. The program compares student answers with the correct answer key and calculates the total marks obtained. This case study demonstrates the use of lists, loops, conditional statements, and counting operations in Python.